Commonly Used Distributions

Lecture 9

Topics in this lecture

- Commonly used discrete distributions
- Commonly used continuous distributions

Discrete Distributions

- Bernoulli
- Binomial
- Geometric
- Poisson

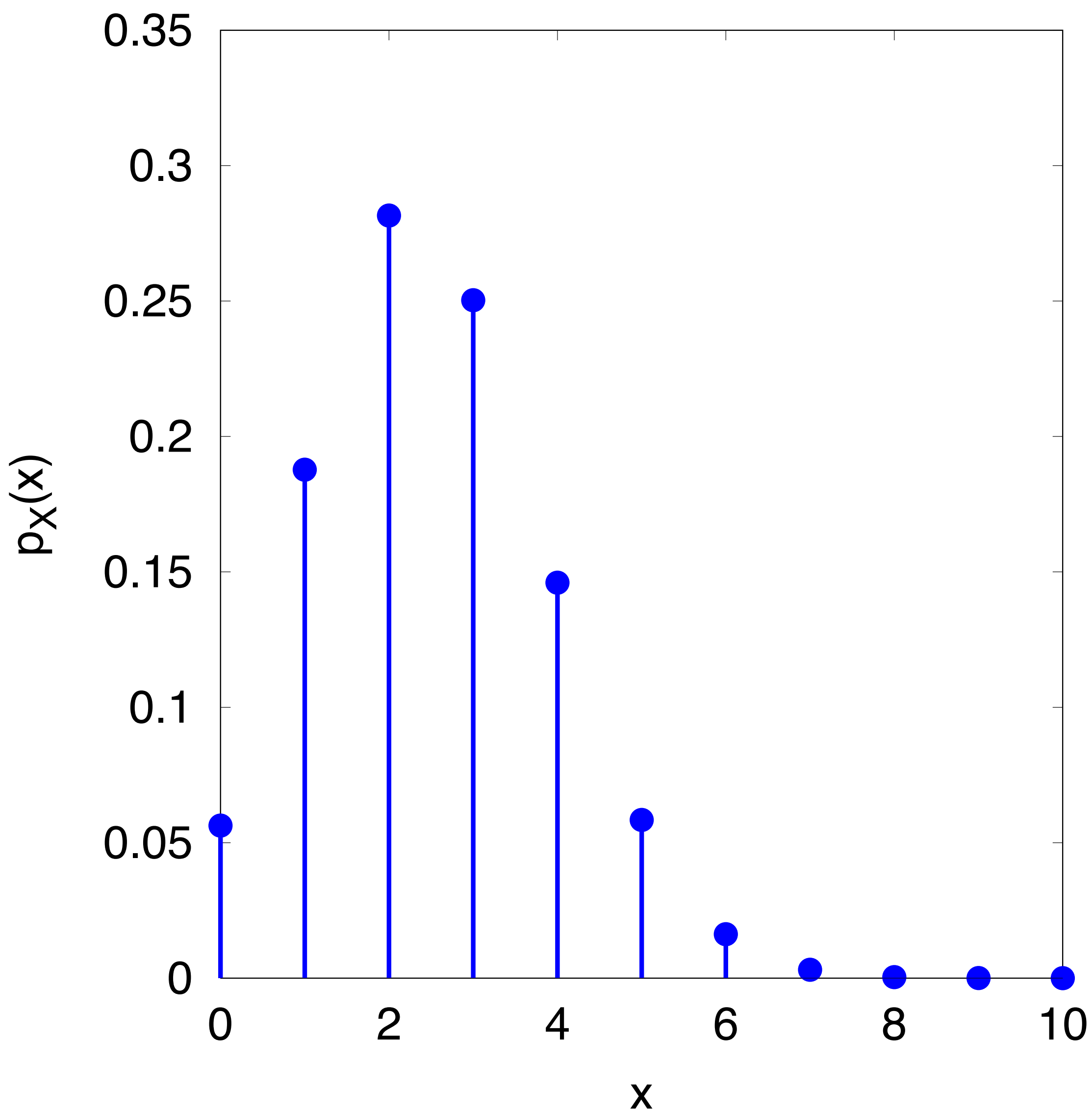
Bernoulli Distribution

Notation	$X \sim \text{Bernoulli}(p)$
PMF	$p_X(x) = \begin{cases} p & , \ x = 1 \\ 1-p & , \ x = 0 \\ 0 & , \ \text{otherwise} \end{cases}$
Mean	$\mu_X = p$
Variance	$\sigma_X^2 = p(1-p)$

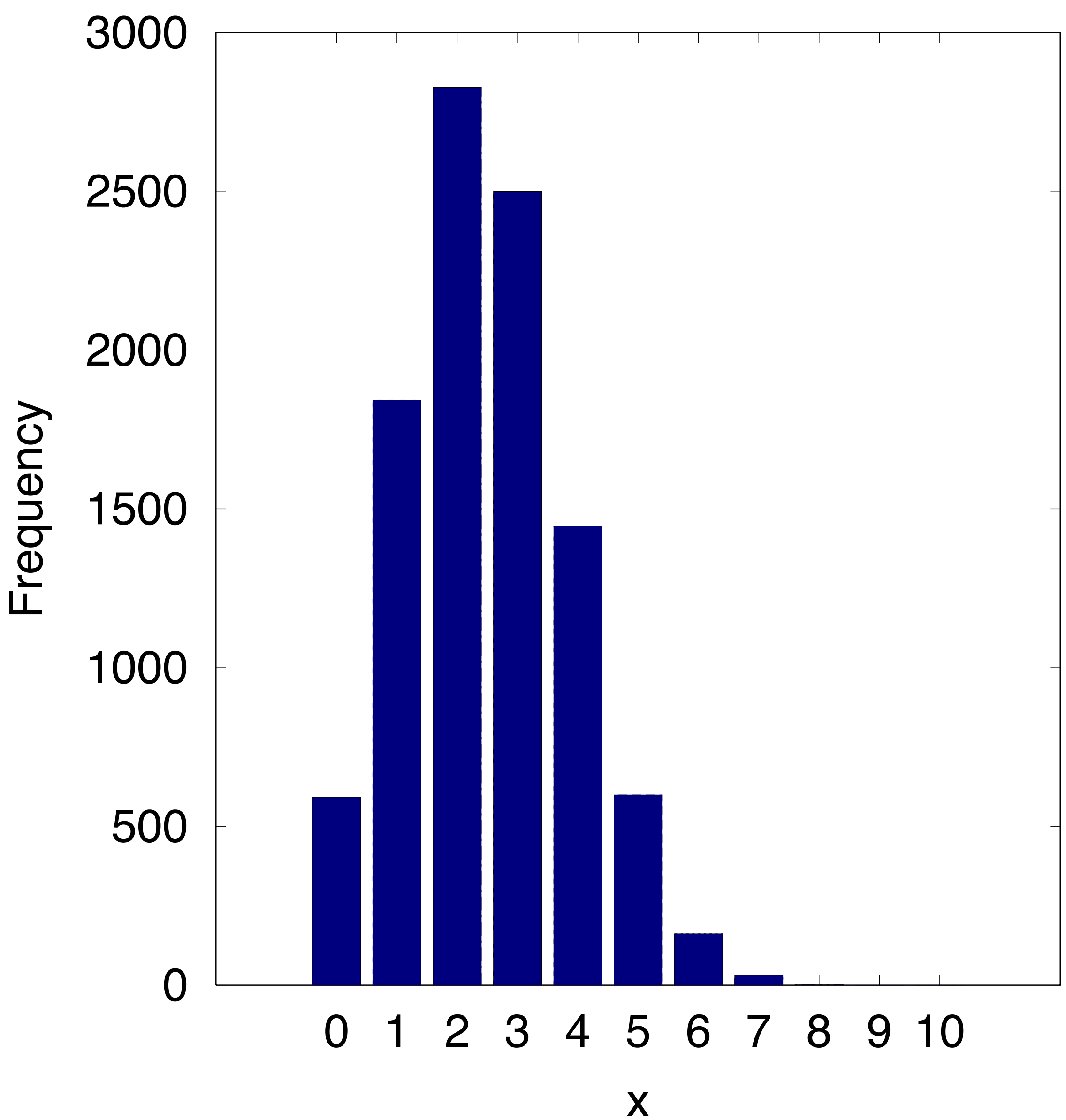
Binomial Distribution

Notation	$X \sim \text{Binomial}(n, p)$
PMF	$p_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & , \ x \in \{0, 1, 2, \ldots, n\} \\ 0 & , \ \text{otherwise} \end{cases}$
Mean	$\mu_X = np$
Variance	$\sigma_X^2 = np(1-p)$

Binomial Distribution



PMF of Binomial(10,0.25)

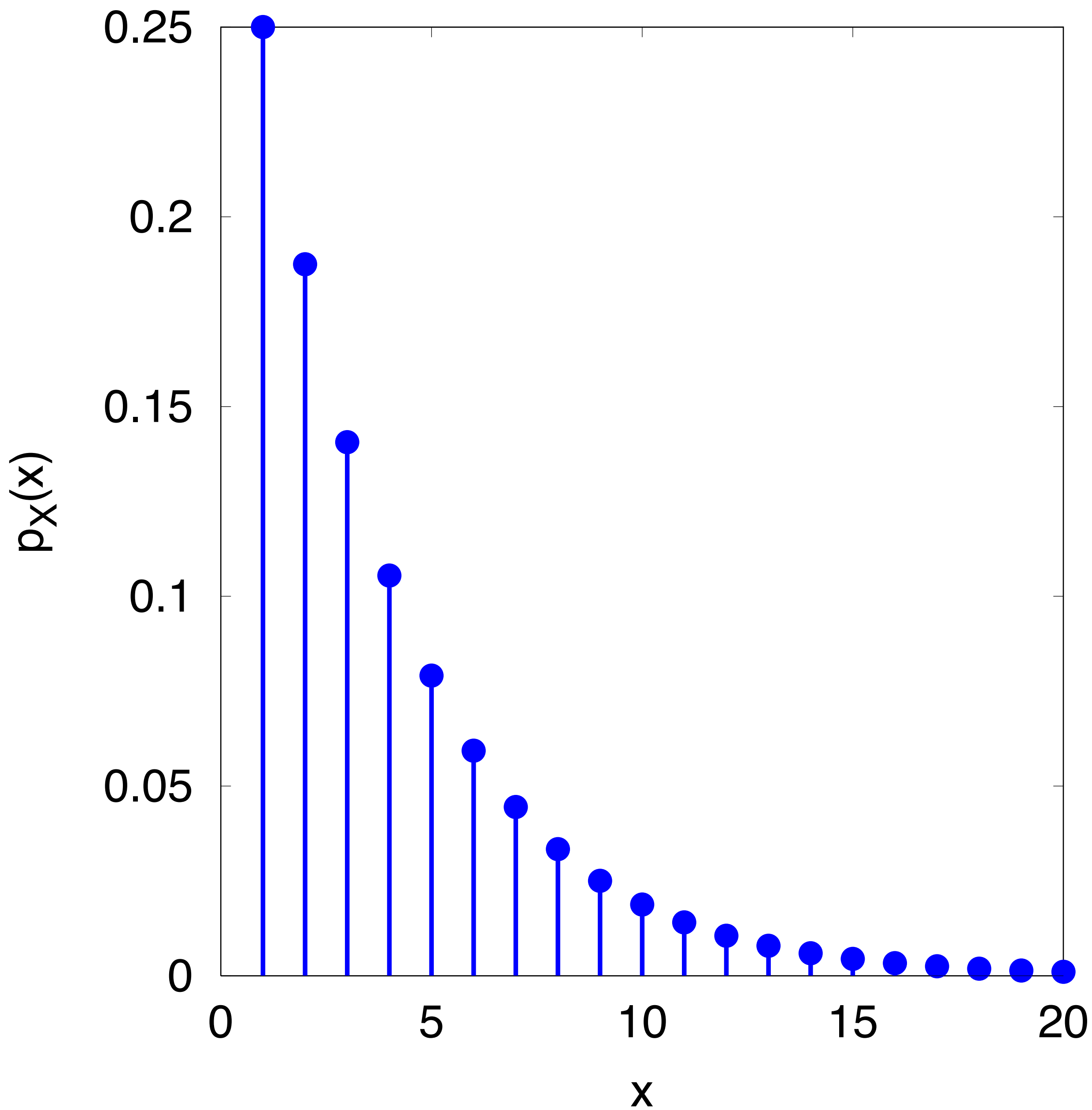


Histogram of samples drawn from Binomial(10,0.25)

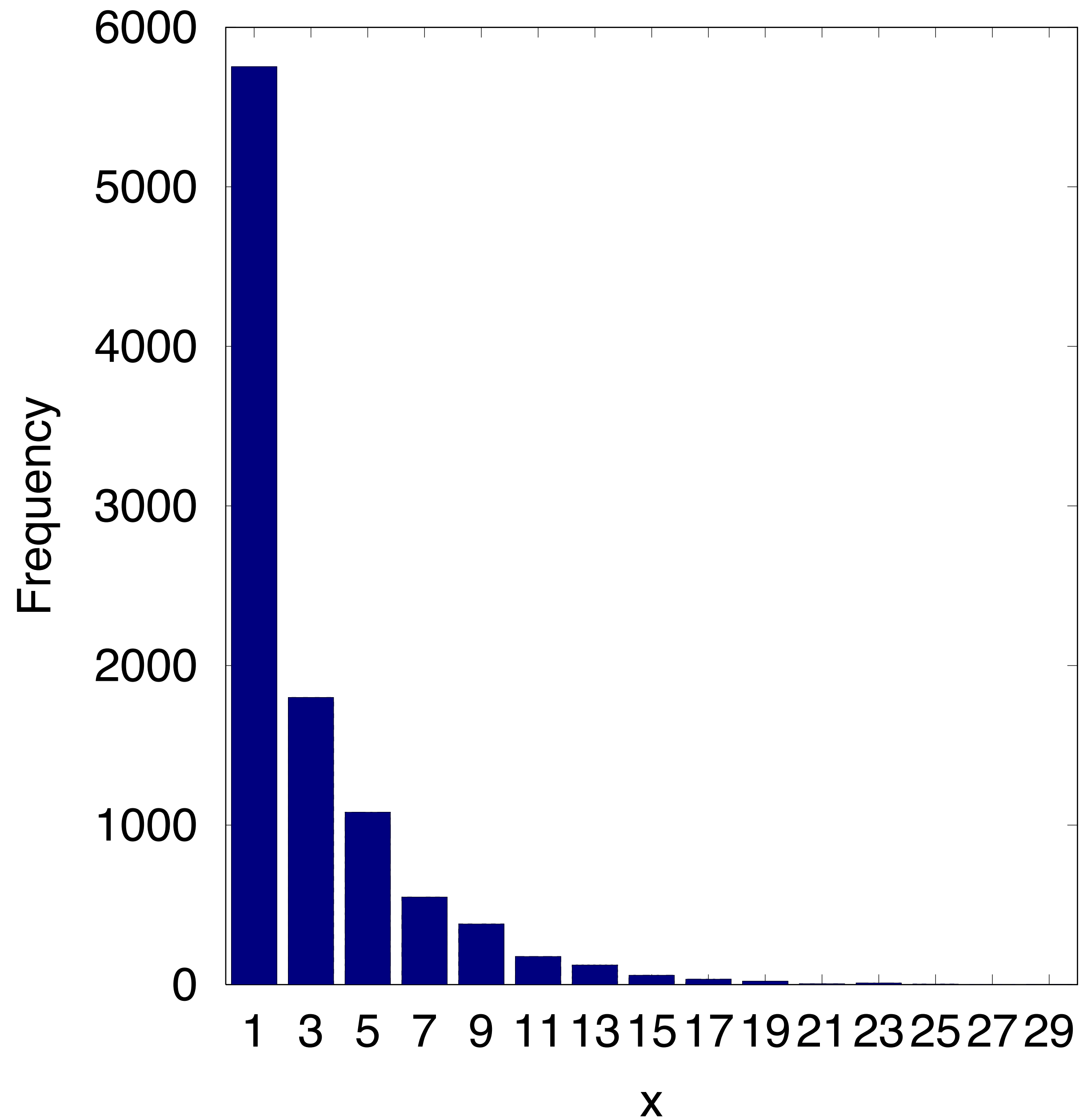
Geometric Distribution

Notation	$X \sim \text{Geometric}(p)$
PMF	$p_X(x) = \begin{cases} p(1-p)^{x-1} & , \ x \in \{1, 2, 3, \dots\} \\ 0 & , \ \text{otherwise} \end{cases}$
Mean	$\mu_X = \dfrac{1}{p}$
Variance	$\sigma_X^2 = \dfrac{1-p}{p^2}$

Geometric Distribution



PMF of Geometric(0.25)

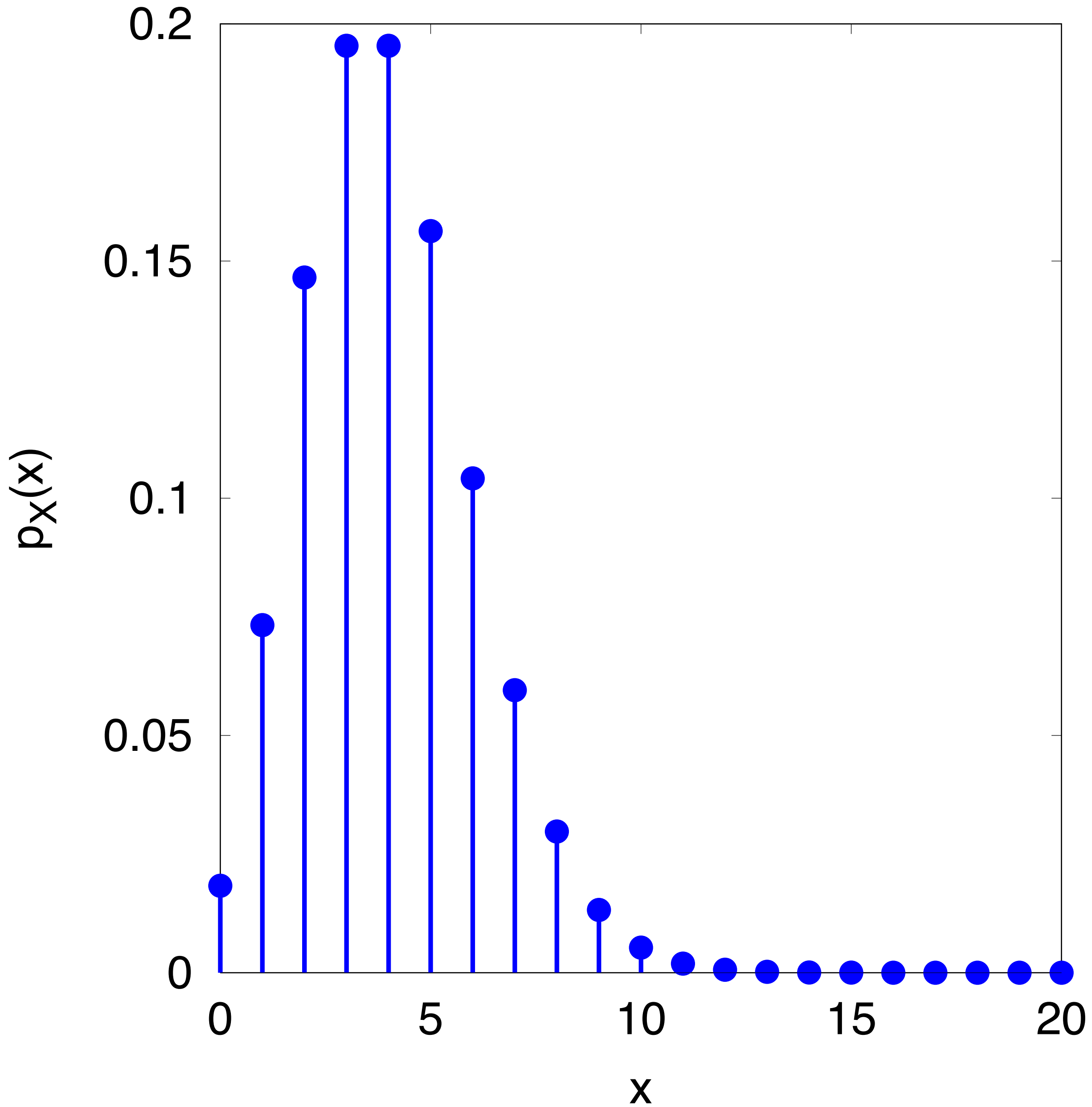


Histogram of samples drawn from Geometric(0.25)

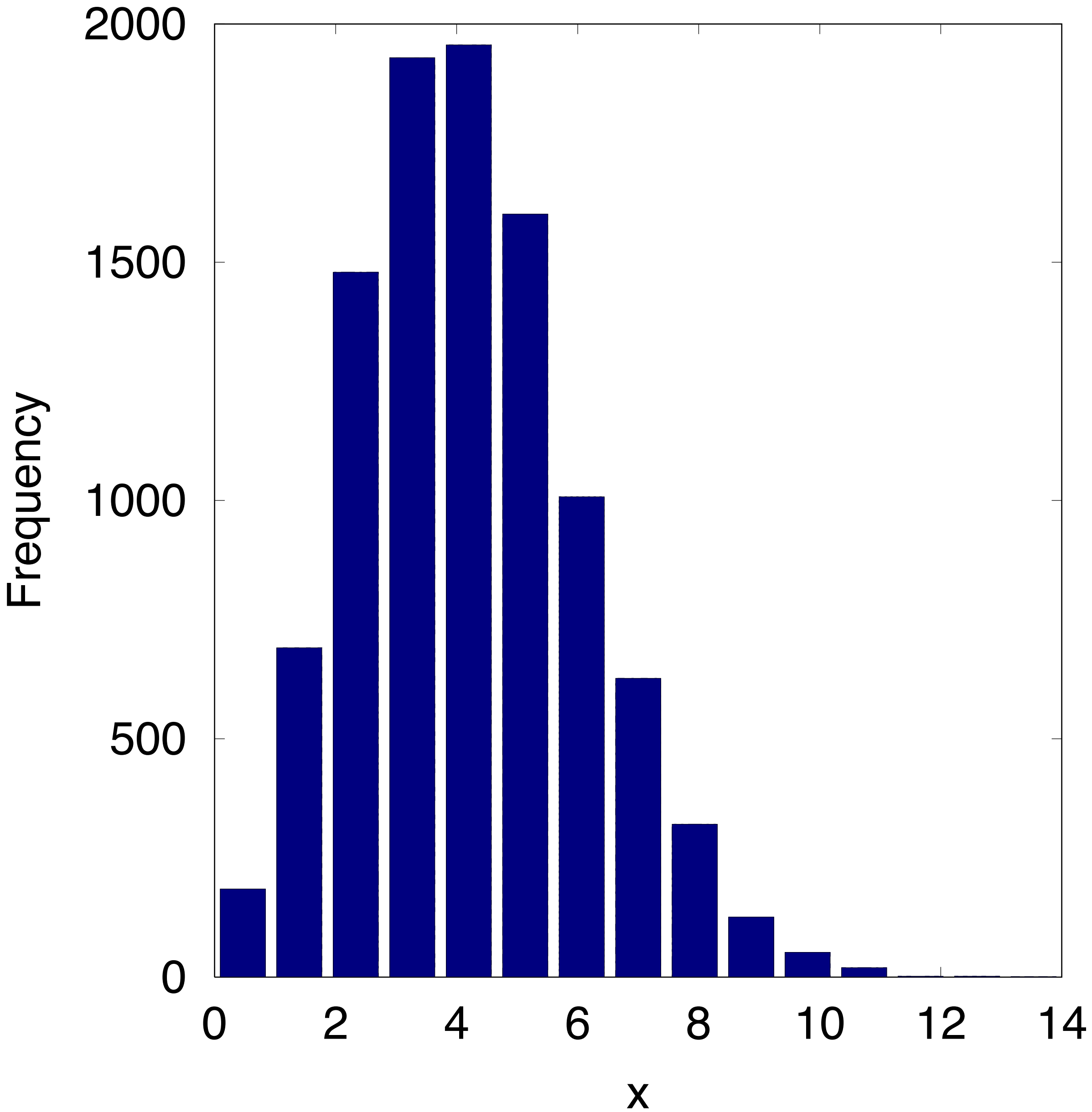
Poisson Distribution

Notation	$X \sim \text{Poisson}(\lambda)$
PMF	$p_X(x) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda} & , \; x \in \{0, 1, 2, \ldots\} \\ 0 & , \; \text{otherwise} \end{cases}$
Mean	$\mu_X = \lambda$
Variance	$\sigma_X^2 = \lambda$

Poisson Distribution



PMF of Poisson(4)



Histogram of samples drawn from Poisson(4)

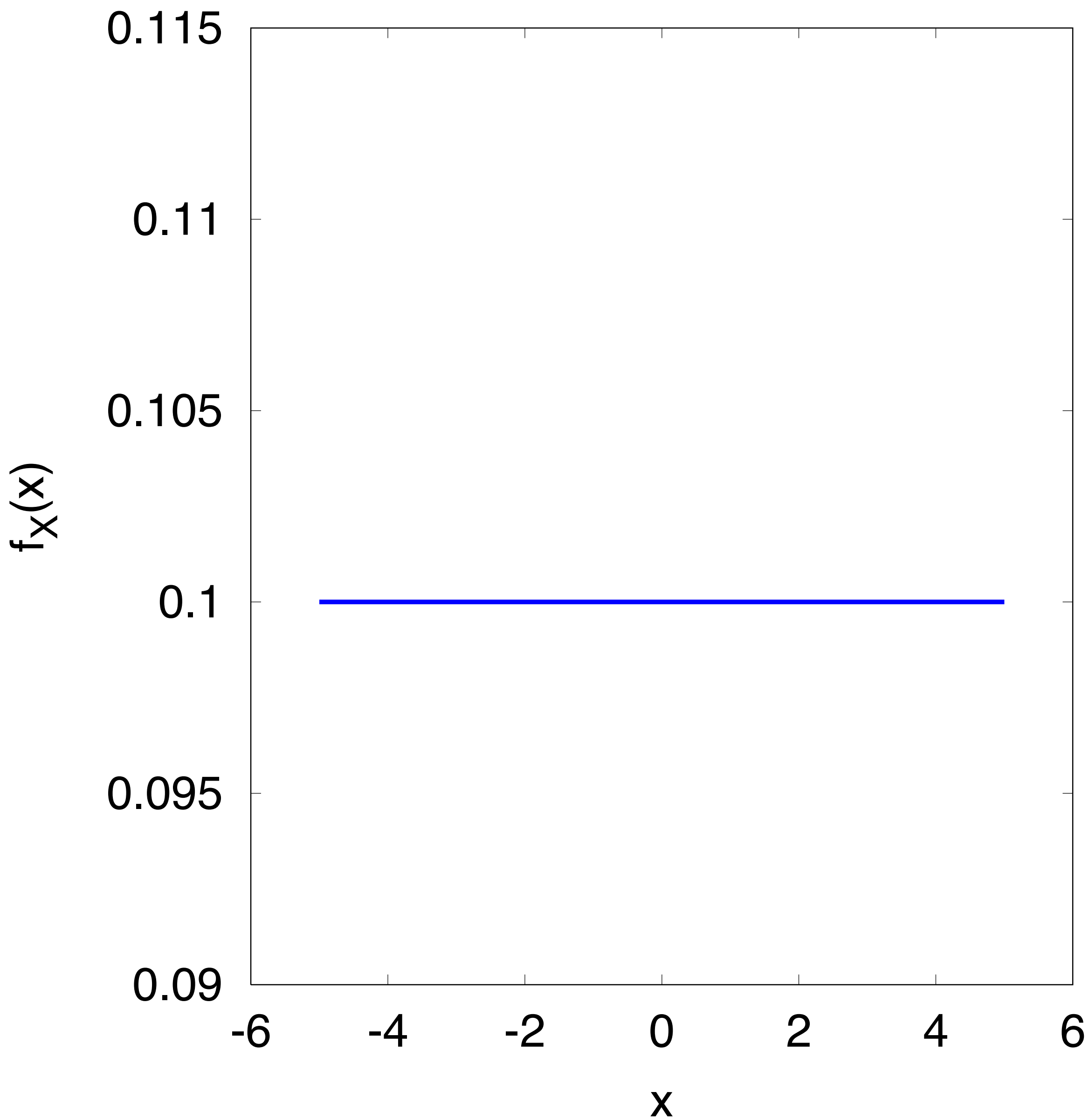
Continuous Distributions

- Uniform
- Exponential
- Normal

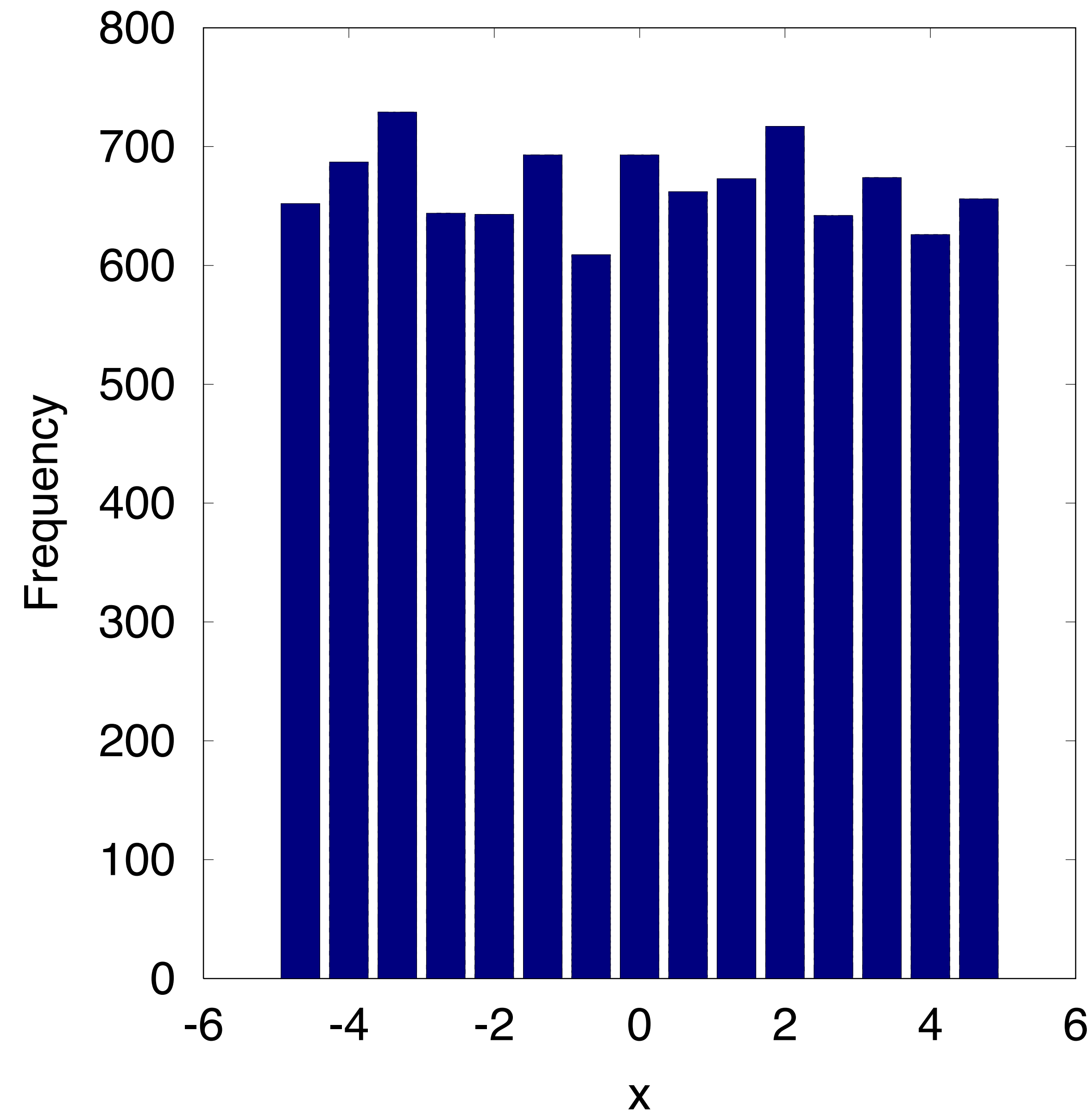
Uniform Distribution

Notation	$X \sim \text{Uniform}(a, b)$
PDF	$f_X(x) = \begin{cases} \frac{1}{b-a} & , \ a < x < b \\ 0 & , \ \text{otherwise} \end{cases}$
Mean	$\mu_X = \dfrac{a+b}{2}$
Variance	$\sigma_X^2 = \dfrac{(b-a)^2}{12}$

Uniform Distribution



PDF of Uniform(-5,5)

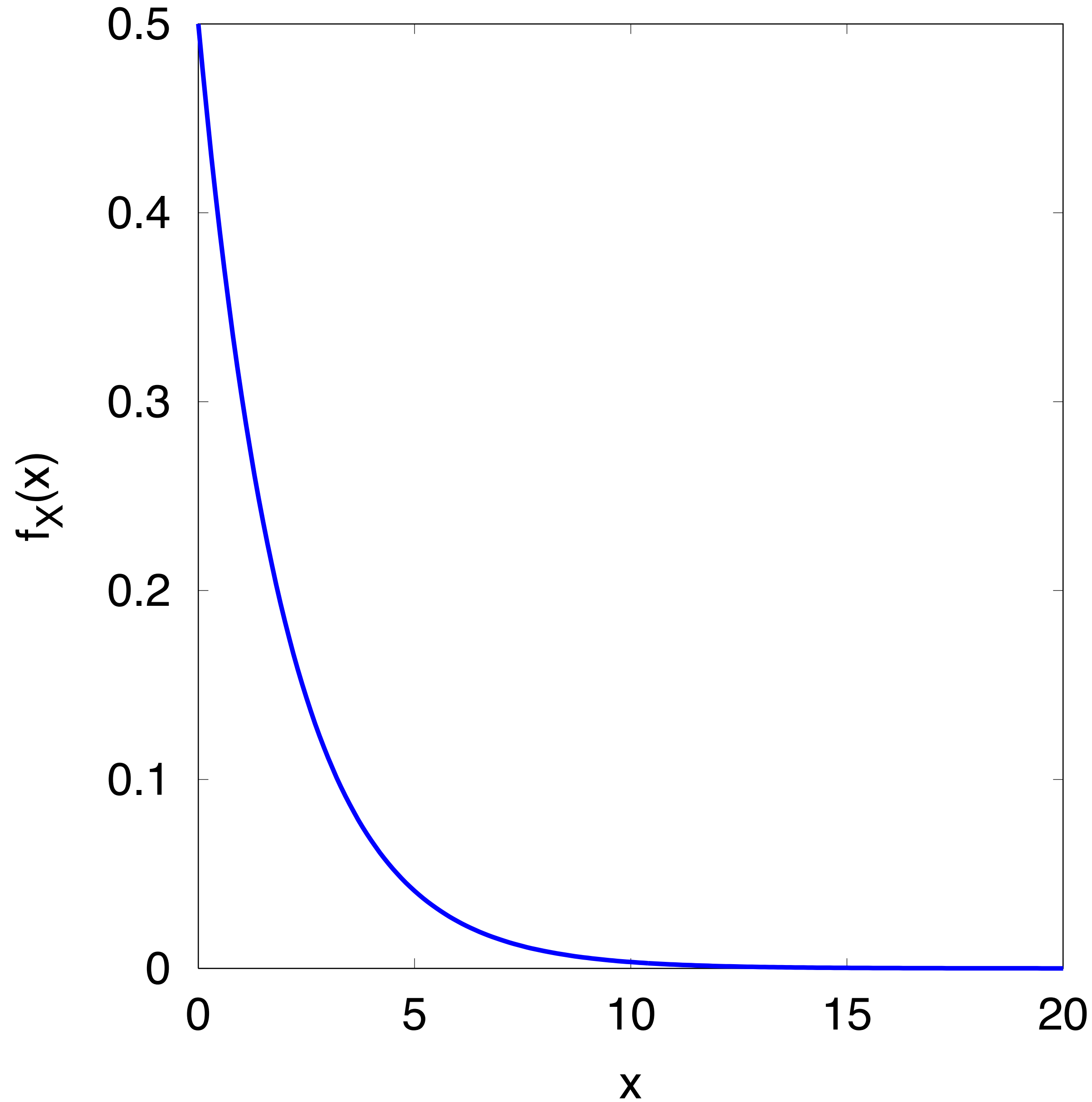


Histogram of samples drawn from Uniform(-5,5)

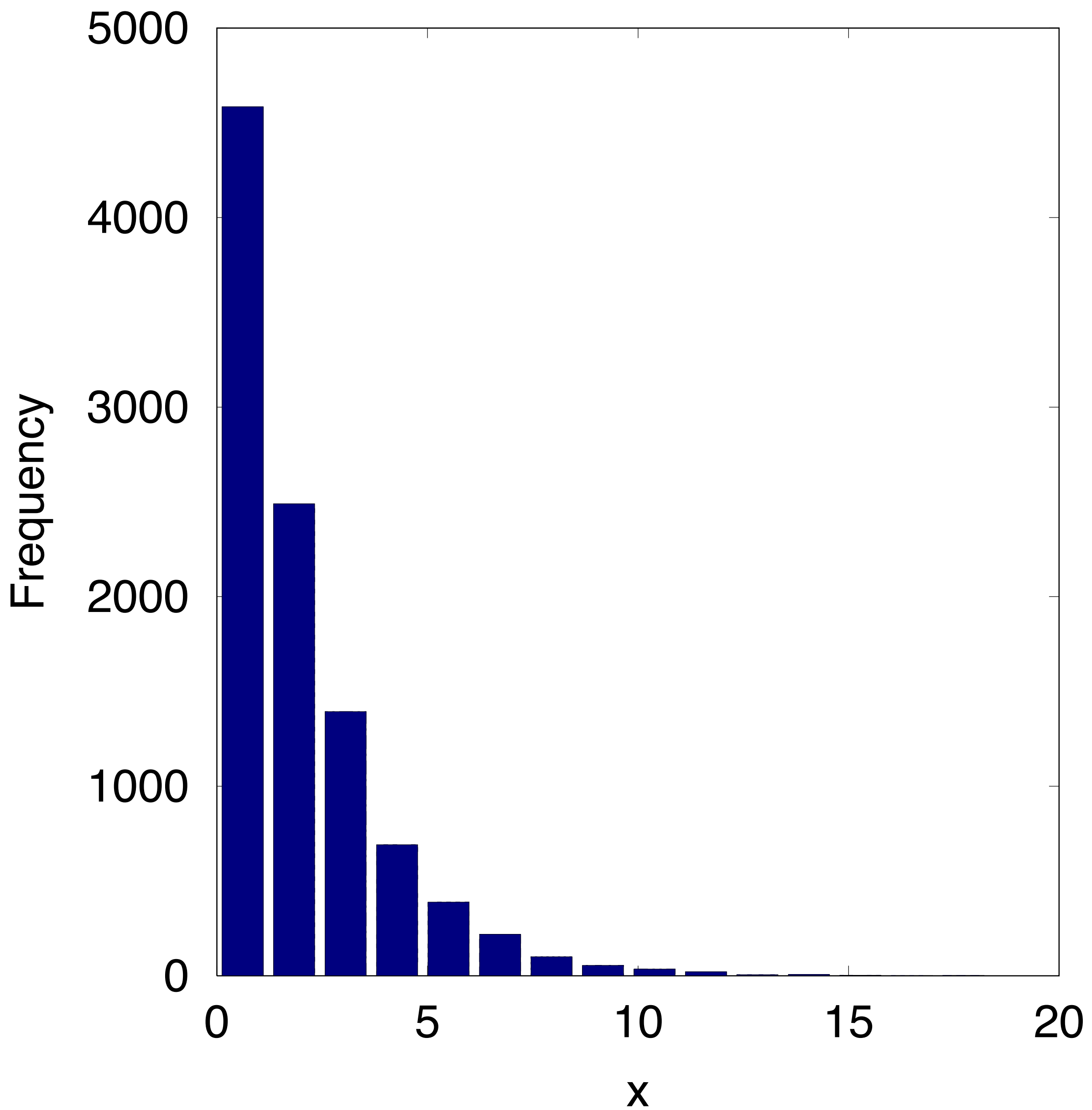
Exponential Distribution

Notation	$X \sim \mathrm{Exp}(\lambda)$
PDF	$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , \ x \geq 0 \\ 0 & , \ \text{otherwise} \end{cases}$
Mean	$\mu_X = \dfrac{1}{\lambda}$
Variance	$\sigma_X^2 = \dfrac{1}{\lambda^2}$

Exponential Distribution



PDF of Exp(2)

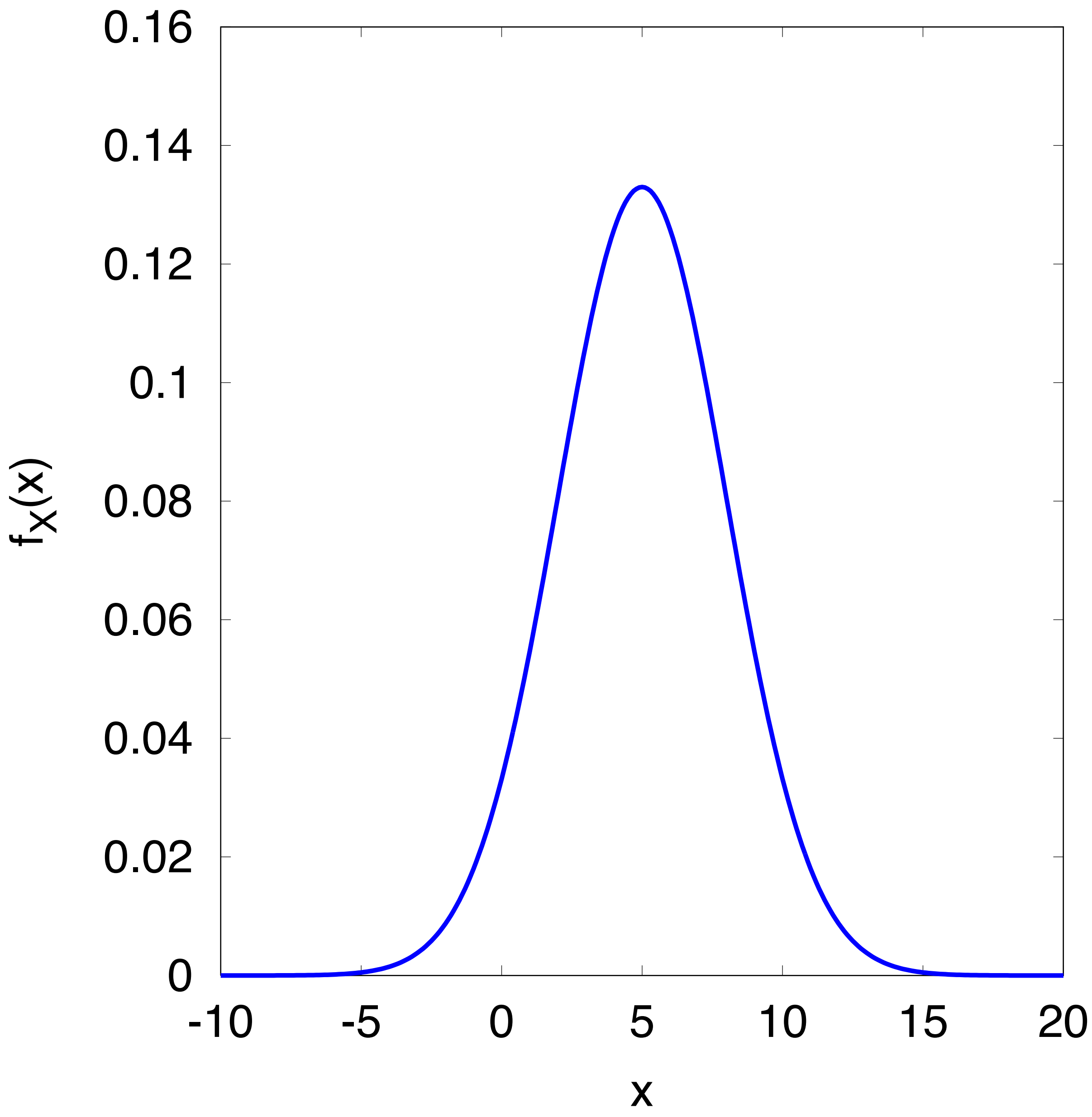


Histogram of samples drawn from Exp(2)

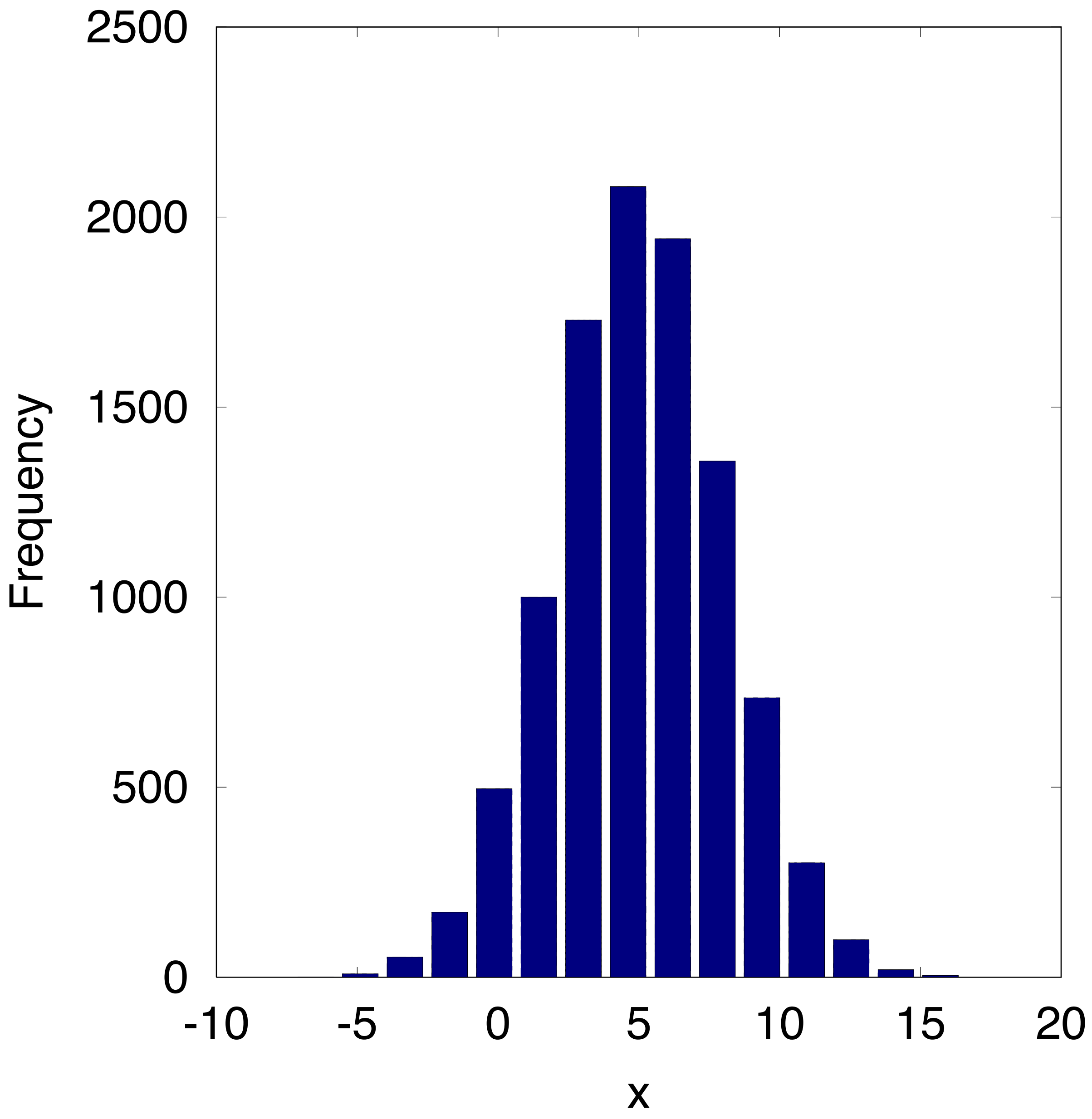
Normal Distribution

Notation	$X \sim \mathcal{N}(\mu, \sigma^2)$
PDF	$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} & , \; -\infty < x < \infty \\ 0 & , \; \text{otherwise} \end{cases}$
Mean	$\mu_X = \mu$
Variance	$\sigma_X^2 = \sigma^2$

Normal Distribution



PDF of N(5,9)



Histogram of samples drawn from N(5,9)

Normal Curve

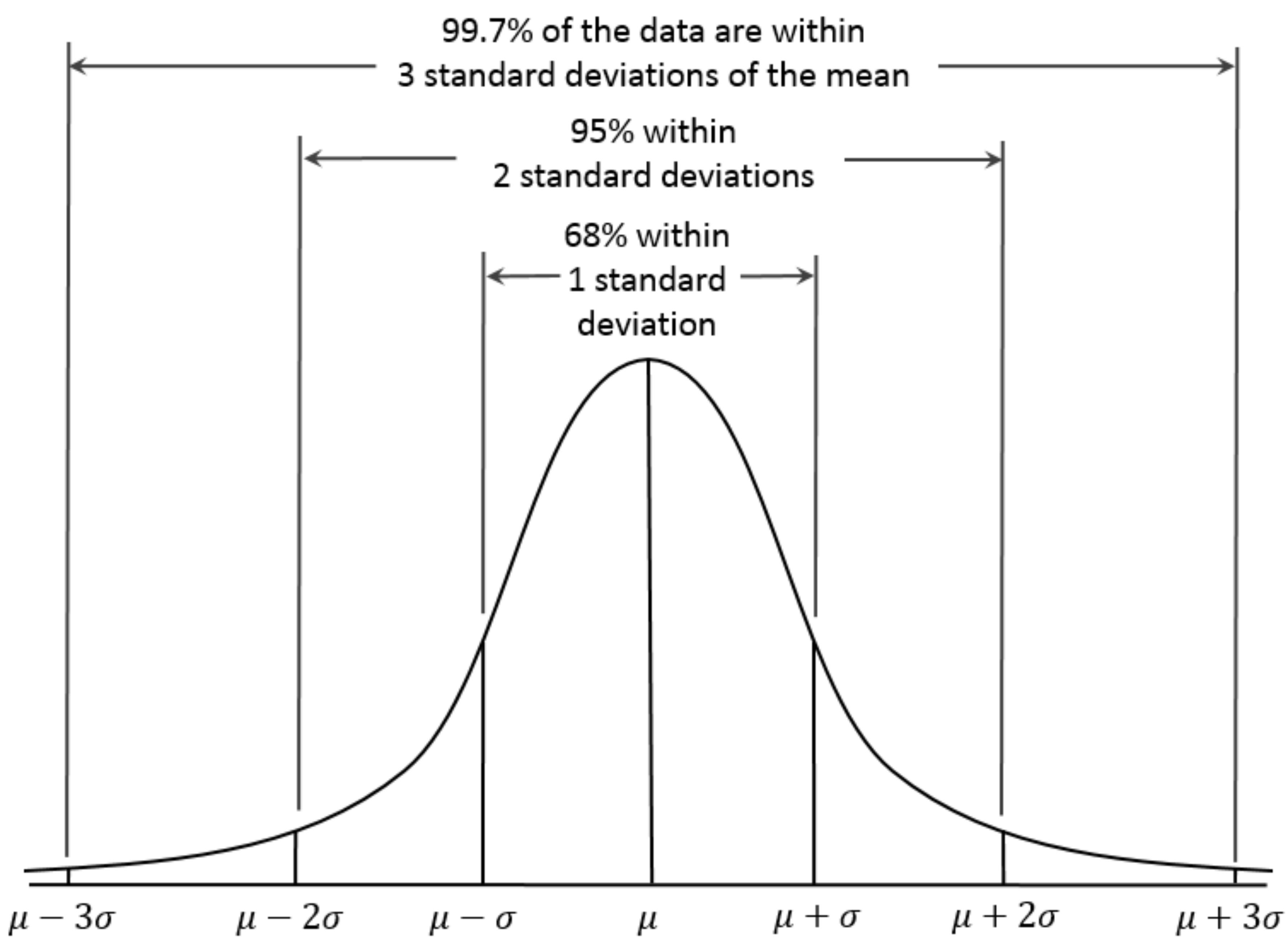


Image Source: Dan Kernler, https://en.wikipedia.org/wiki/File:Empirical_Rule.PNG

Standard Normal

- Normal population with mean 0 and standard deviation 1

- Converting normal population with mean μ and standard deviation σ to a standard normal population

$$z = \frac{x - \mu}{\sigma}$$

- This value is called the "z-score" of x

Standard Normal Table

- For standard normal

$$P(Z \leq z) = \Phi(z)$$

- For $X \sim$ Normal(μ, σ^2)

$$P(X \leq x) = P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right)$$
$$= P\left(Z \leq \frac{x-\mu}{\sigma}\right)$$
$$= \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Source:

http://www.stat.ufl.edu/~athienit/Tables/Ztable.pdf

Standard Normal Probabilities

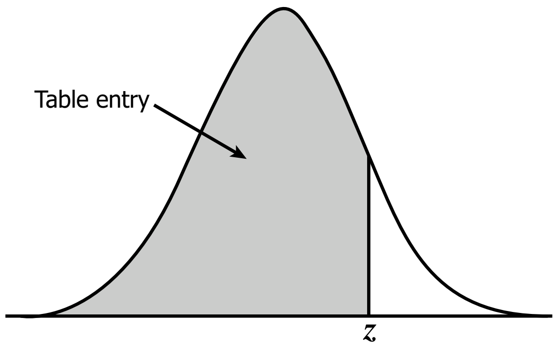


Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916

Example 1

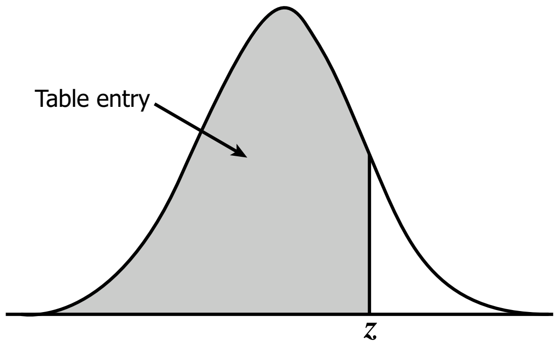
- Aluminum sheets used to make beverage cans have thickness (in mm) that are normally distributed with mean 10 and standard deviation 1.3. A particular sheet is 10.8 mm thick. Find the z-score

- The thickness of a certain sheet has a z-score of -1.7. Find the thickness of the sheet in the original units of mm.

Example 2

- Find the area under the normal curve to the left of z = 0.47

Standard Normal Probabilities



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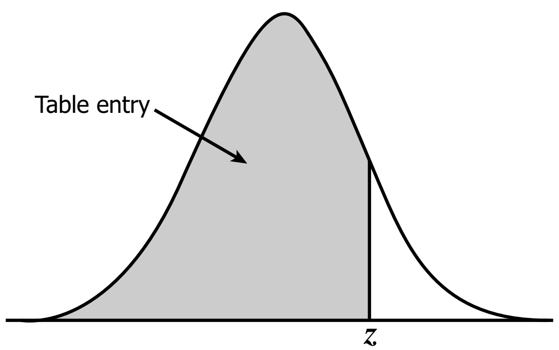
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 2

- Find the area under the normal curve to the left of z = -0.35

Standard Normal Probabilities



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0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
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0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
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1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 2

- Find the area under the normal curve to the right of z = 1.38

Standard Normal Probabilities

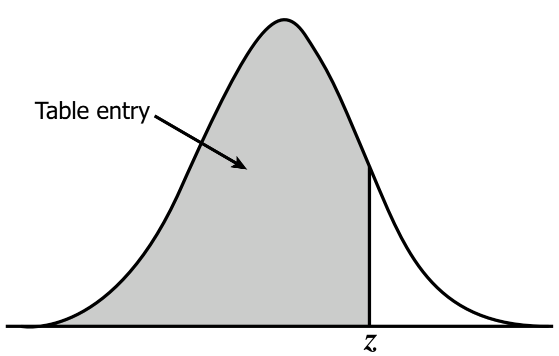

Table entry for z is the area under the standard normal curve to the left of z.

z	**.00**	**.01**	**.02**	**.03**	**.04**	**.05**	**.06**	**.07**	**.08**	**.09**
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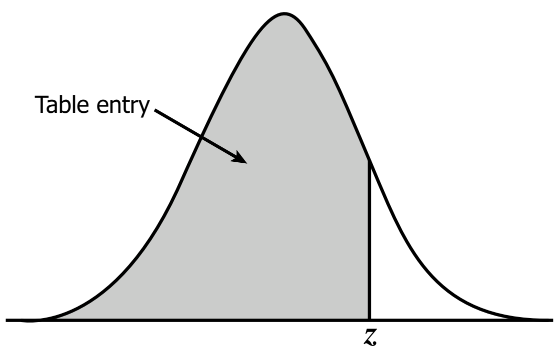
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 2

- Find the area under the normal curve to between z = 0.71 and z = 1.28

Standard Normal Probabilities



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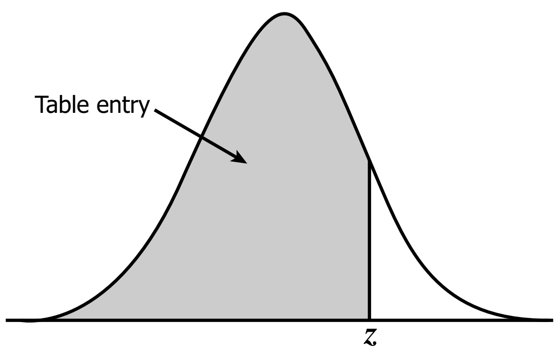
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 3

- What z-score corresponds to the 75th percentile, the 25th percentile, and the median?

Standard Normal Probabilities



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0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 4

- Lifetime of batteries in a certain application are normally distributed with mean 50 hours and standard deviation 5 hours. Find the probability that a randomly chosen battery lasts between 42 and 52 hours

Standard Normal Probabilities

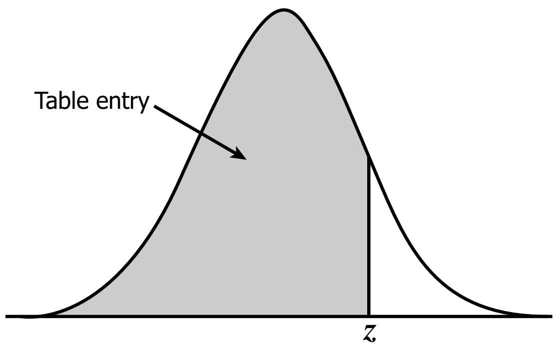

Table entry for z is the area und
to the left of z.

z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686

Example 5

- A process manufactures balls whose diameters are normally distributed with mean 2.50 cm and standard deviation 0.008 cm. Specifications call for the diameter to be in the interval 2.5±0.01 cm. What proportion of the balls will meet the specification?

Standard Normal Probabilities

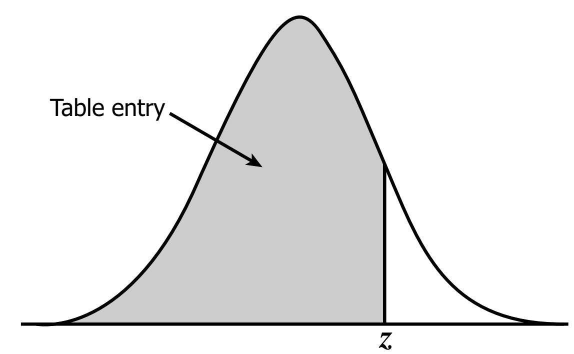

Table entry for z is the area under the
to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.527
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.567
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.606
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.644
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.680
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.715
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.748
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.779
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.807
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.834
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.857
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.879
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.898
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.914
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.929
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.941